

Project (1) Report

Title:

Network Penetration Testing

Type:

Security Assessment Lab

Submitted By:

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1. Introduction

Penetration testing is a cybersecurity practice used to identify security vulnerabilities in computer systems and networks by simulating real-world attacks in a controlled environment. It helps security professionals evaluate the effectiveness of existing security measures and discover potential weaknesses before they can be exploited by attackers.

Network penetration testing focuses on identifying active hosts, open ports, running services, and potential security risks within a network. In this project, **Kali Linux** was used as the attacker machine and **Metasploitable 2** as the target system. **Nmap** was used for network scanning and service enumeration, while **Wireshark** was used to capture and analyze network traffic during the assessment.

2. Objectives

The main objectives of this project are:

- To set up a penetration testing lab using Kali Linux and Metasploitable 2.
- To perform network scanning using Nmap.
- To identify active hosts, open ports, and running services.
- To capture and analyze network traffic using Wireshark.
- To assess the security posture of the target system.
- To document the findings and provide security recommendations.

3. Lab Environment

The penetration testing lab was created in a virtual environment using VirtualBox. Kali Linux was configured as the attacker machine, while Metasploitable 2 was used as the intentionally vulnerable target machine. Nmap was used to perform network scanning and service enumeration, and Wireshark was used to capture and analyze network traffic during the assessment.

Component	Details
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Virtualization Platform	Oracle VirtualBox
Network Scanner	Nmap 7.99

Packet Analyzer	Wireshark
Web Browser	Firefox
Attacker IP Address	192.168.1.4
Target IP Address	192.168.1.3

4. Network Configuration

Both virtual machines were connected to the same VirtualBox network to enable communication between the attacker and target systems. Connectivity was verified using the `ping` command before performing the penetration testing activities.

Kali Linux (Attacker)

IP: 192.168.1.4

|
|

Metasploitable 2 (Target)

IP: 192.168.1.3

5. Tools Used

Tool	Purpose
Kali Linux	Used as the attacker machine to perform penetration testing.
Nmap	Used to discover hosts, scan open ports, and identify running services.
Wireshark	Used to capture and analyze network traffic during the assessment.
Firefox	Used to access and inspect the target's web services.

6. Methodology

The following methodology was followed to complete the network penetration testing project:

Install Kali Linux

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Install Metasploitable 2

Configure Virtual Network

Verify Connectivity (Ping)

Basic Nmap Scan

Service Enumeration (-sV)

Aggressive Scan (-A)

Capture Traffic (Wireshark)

Analyze Results

Document Findings & Generate Report

7. Reconnaissance

Reconnaissance is the first phase of penetration testing in which information about the target system is collected. In this project, Nmap was used to identify active hosts, open ports, running services, and operating system information before further analysis.

7.1 Connectivity Test (Ping)

Command:

ping 192.168.1.3

Screenshot:

```
(aliza@kali)-[~]
└─$ ping 192.168.1.3
PING 192.168.1.3 (192.168.1.3) 56(84) bytes of data:
64 bytes from 192.168.1.3: icmp_seq=1 ttl=64 time=4.00 ms
64 bytes from 192.168.1.3: icmp_seq=2 ttl=64 time=2.57 ms
64 bytes from 192.168.1.3: icmp_seq=3 ttl=64 time=1.78 ms
64 bytes from 192.168.1.3: icmp_seq=4 ttl=64 time=2.25 ms
64 bytes from 192.168.1.3: icmp_seq=5 ttl=64 time=2.43 ms
64 bytes from 192.168.1.3: icmp_seq=6 ttl=64 time=3.01 ms
64 bytes from 192.168.1.3: icmp_seq=7 ttl=64 time=3.15 ms
64 bytes from 192.168.1.3: icmp_seq=8 ttl=64 time=3.53 ms
64 bytes from 192.168.1.3: icmp_seq=9 ttl=64 time=2.75 ms
64 bytes from 192.168.1.3: icmp_seq=10 ttl=64 time=6.30 ms
64 bytes from 192.168.1.3: icmp_seq=11 ttl=64 time=2.07 ms
64 bytes from 192.168.1.3: icmp_seq=12 ttl=64 time=2.22 ms
64 bytes from 192.168.1.3: icmp_seq=13 ttl=64 time=2.89 ms
64 bytes from 192.168.1.3: icmp_seq=14 ttl=64 time=3.13 ms
64 bytes from 192.168.1.3: icmp_seq=15 ttl=64 time=2.23 ms
^C
--- 192.168.1.3 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14608ms
rtt min/avg/max/mdev = 1.784/2.954/6.297/1.059 ms
(aliza@kali)-[~]
```

Observation:

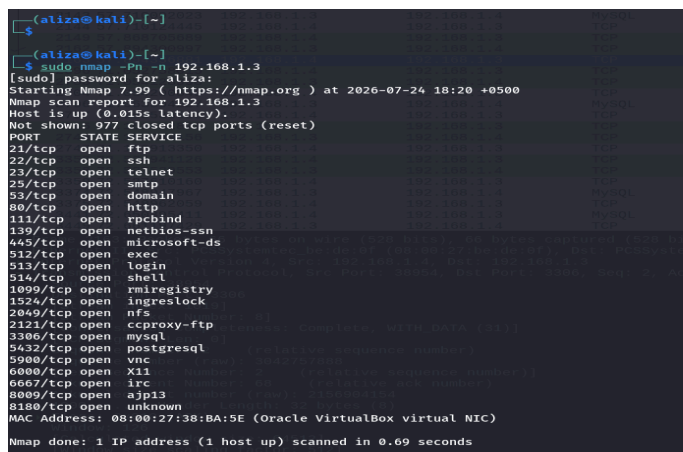
The **ping** command was used to verify connectivity between the Kali Linux machine and the Metasploitable 2 target. The successful ICMP Echo Reply packets confirmed that the target system was reachable and ready for further network scanning.

7.2 Basic Nmap Scan

Command:

```
sudo nmap -Pn -n 192.168.1.3
```

Screenshot:



```
(aliza@kali)-[~]
└─$ sudo nmap -Pn -n 192.168.1.3
[sudo] password for aliza:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-24 18:20 +0500
Nmap scan report for 192.168.1.3
Host is up (0.015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:38:BA:5E (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
```

Observation:

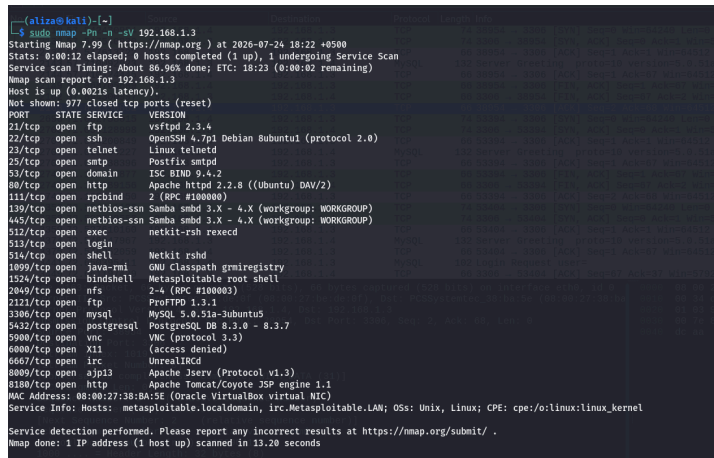
The basic Nmap scan confirmed that the target host was online and identified multiple open TCP ports. Several services, including FTP, SSH, Telnet, HTTP, SMB, MySQL, and PostgreSQL, were found running on the target system, indicating a large attack surface.

7.3 Service Version Scan

Command:

```
sudo nmap -Pn -n -sV 192.168.1.3
```

Screenshot:



```

--(alixe@kali)-[~]
LL$ sudo nmap -Pn -n -sV 192.168.1.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-24 18:22 +0500
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 86.90s done; ETC: 18:23 (0:00:02 remaining)
Nmap scan report for 192.168.1.3
Host is up (0.0021s latency).
Not shown: 577 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
113/tcp   open  epobind      2 (RPC #180000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        Netkit rshd
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath gmicregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2+ (RPC #180000)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-Jubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.9 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8080/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:38:BA:5E (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.20 seconds
```

Observation:

The service version scan identified the software and versions running on the open ports. The scan detected:

- **FTP** – vsftpd 2.3.4 and ProFTPD 1.3.1
- **SSH** – OpenSSH 4.7p1
- **HTTP** – Apache HTTP Server 2.2.8
- **SMB** – Samba 3.0.20
- **MySQL** – MySQL 5.0.51a
- **PostgreSQL** – PostgreSQL 8.3.x
- **Tomcat** – Apache Tomcat 5.5

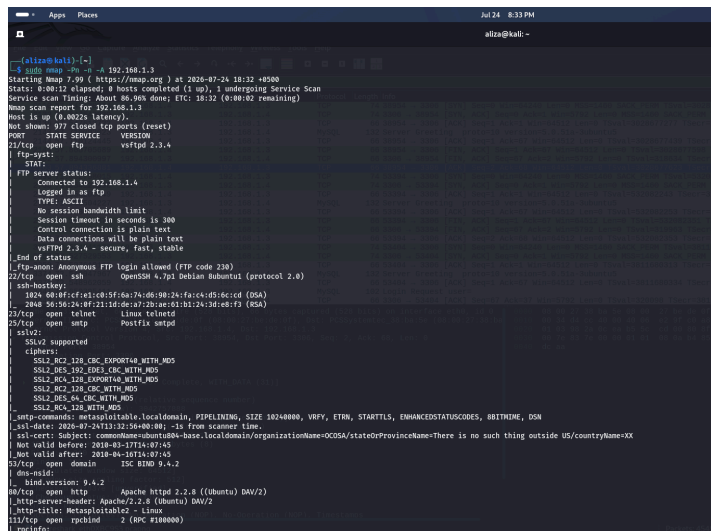
This information helps identify outdated software that may contain known vulnerabilities.

7.4 Aggressive Nmap Scan

Command:

```
sudo nmap -Pn -n -A 192.168.1.3
```

Screenshot:



```
alisha@kali:~$ sudo nmap -Pn -n -A 192.168.1.3
Starting Nmap 7.99 ( https://nmap.org ) at 2020-07-26 18:32 +0500
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 86.9% done; ETC: 18:20 (0:00:02 remaining)
Nmap scan report for 192.168.1.3
Host is up (0.002s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp             vsftpd 2.3.4
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.1.4
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_vsFTPd 2.3.4 - secure, fast, stable
|_End of status.
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh             OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 66:f1cf:01:00:f5:6a:7a:de:90:2a:fa:c4:d5:6c:ed (RSA)
|_  2048 56:56:24:f8:21:1d:de:a7:2b:ae:01:b1:24:3d:ed:f3 (RSA)
23/tcp    open  telnet          Linux telnetd
25/tcp    open  smtp            Postfix smtpd
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_  SSLv2_KCZ_128_CK_EXPORT40_WITH_MD5
|_  SSLv2_DG5_192_DG3_CK_WITH_MD5
|_  SSLv2_KCZ_128_EXPORT40_WITH_MD5
|_  SSLv2_KCZ_128_EXPORT40_WITH_MD5
|_  SSLv2_KCZ_128_CK_WITH_MD5
|_  SSLv2_DG5_64_CK_WITH_MD5
|_  SSLv2_KCZ_128_WITH_MD5
|_smb-combats: Metasploitable.localdomain, PIPELINING, SIZE 16244800, VFPV, ETRN, STARTTLS, ENHANCEDSTATUSCODES, HBITHINE, DSN
|_smb-date: 2020-07-24T13:32:56+00:00 -1s from scanner time.
|_smb-err: Subject: commonName=mountb-base.localdomain/organizationName=OC09A/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_  Not valid before: 2018-03-17T14:07:45
|_  Not valid after: 2019-04-16T14:07:45
25/tcp    open  domain         ISC BIND 9.4.2
|_dns-nsid:
|_  bind-version: 9.4.2
80/tcp    open  http            Apache/2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_22/tcp    open  rsh             2 (RCP #100000)
|_ncinfo:
```

Observation:

The aggressive scan provided detailed information about the target system, including operating system detection, service versions, default Nmap Scripting Engine (NSE) results, and traceroute information. The scan identified the target as a Linux-based system, detected service banners, collected additional security-related information, and confirmed that the target was one network hop away from the attacker machine.

Information Obtained

- **Operating System Detection:** Linux 2.6.x
- **Version Detection:** Identified versions of FTP, SSH, Apache, Samba, MySQL, PostgreSQL, and Tomcat services.
- **NSE Script Results:** Displayed additional information such as anonymous FTP access, SMB security settings, SSL certificate details, and service configurations.
- **Traceroute:** Confirmed the target was reachable in **1 network hop**, indicating both virtual machines were connected to the same local network.

8. Wireshark Analysis

Wireshark was used to capture and analyze the network traffic generated during the penetration testing process. Different protocol filters were applied to observe communication between the Kali Linux and Metasploitable 2 virtual machines.

8.1 ARP Analysis

Filter:

arp

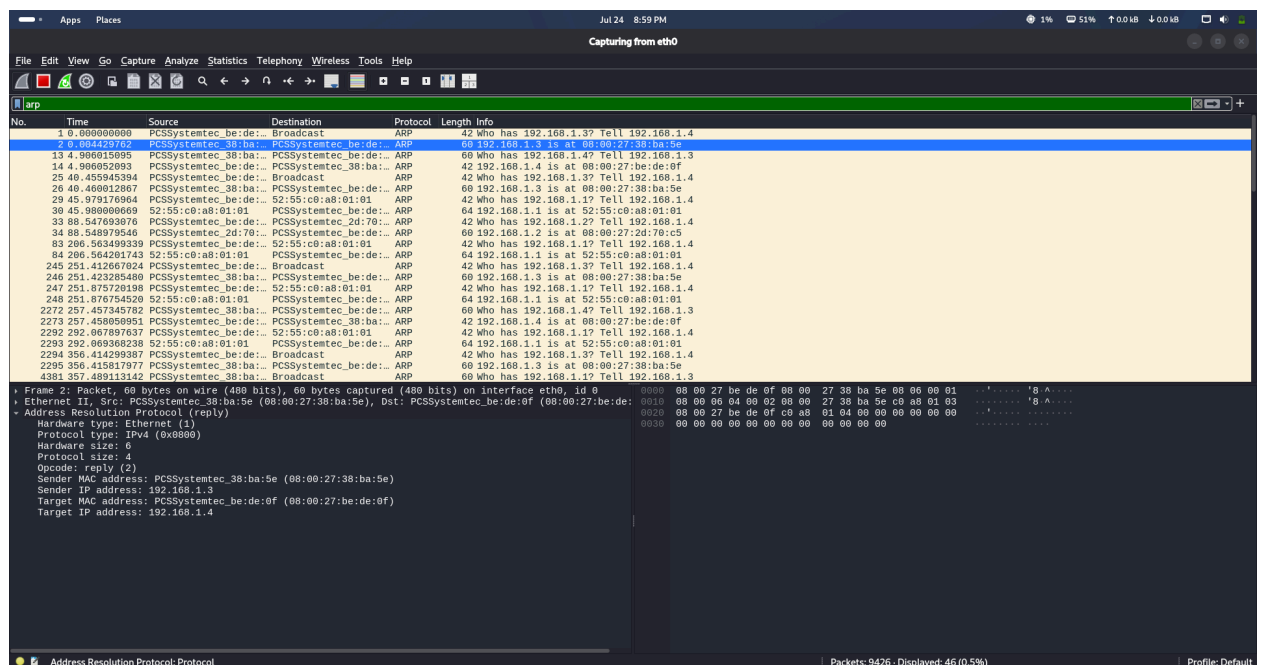
Purpose:

To observe the Address Resolution Protocol (ARP) used to resolve IP addresses into MAC addresses before communication begins.

Observation:

ARP Request and ARP Reply packets were successfully captured, confirming that the attacker machine resolved the target's MAC address before transmitting network traffic.

Screenshot:



8.2 ICMP Analysis

Filter:

icmp

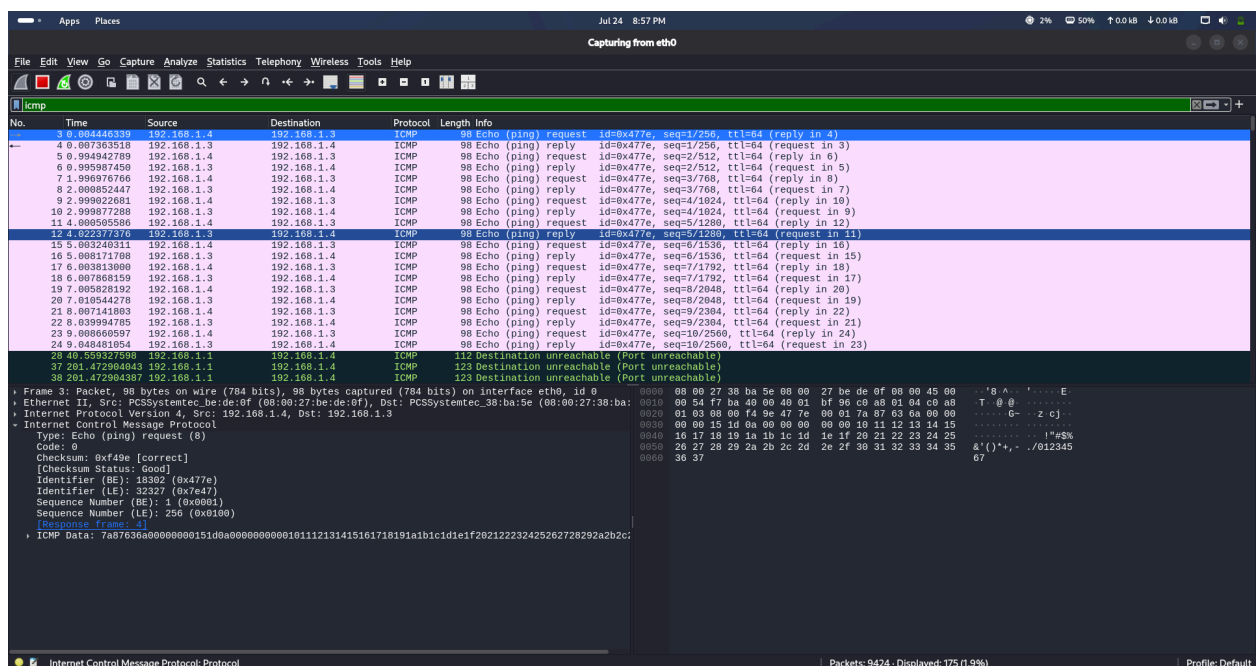
Purpose:

To monitor ICMP packets used for network connectivity testing.

Observation:

ICMP Echo Request and Echo Reply packets were captured during the ping test, confirming successful communication between the attacker and target systems. ICMP Destination Unreachable messages were also observed during scanning, which is normal when probing unavailable services or ports.

Screenshot:



8.3 TCP Analysis

Filter:

tcp.flags.syn == 1 || tcp.flags.ack == 1

Purpose:

To analyze the TCP connection establishment process between the attacker and the target.

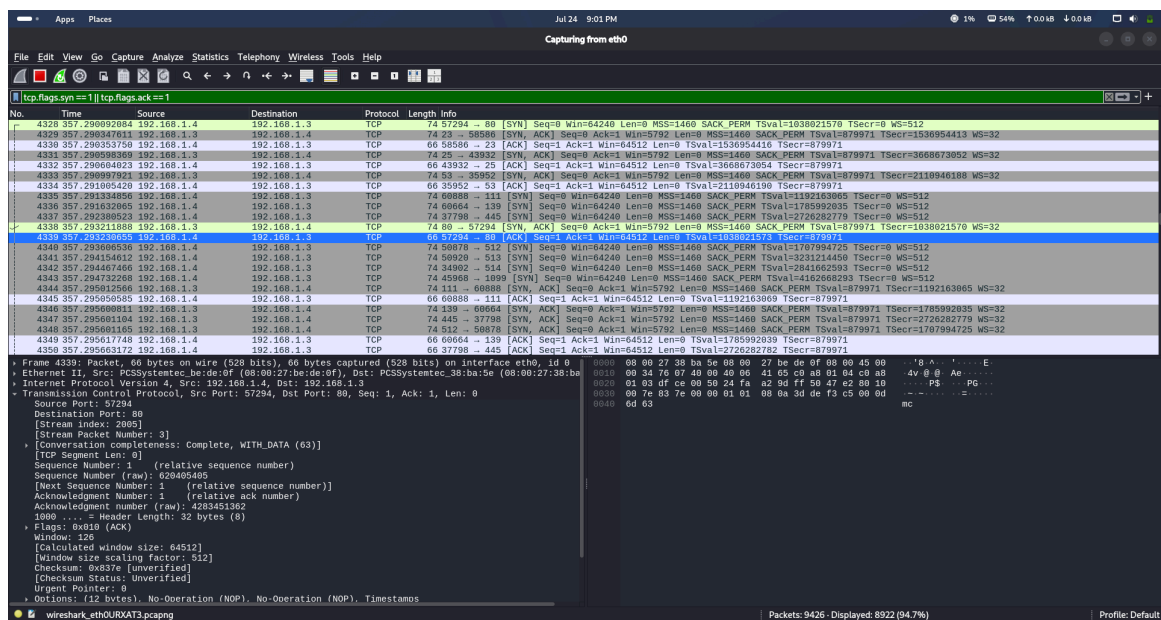
Observation:

The captured packets show the TCP Three-Way Handshake, which establishes a reliable connection before data transmission. The sequence observed was:

- **SYN** – The client requests a connection.
- **SYN-ACK** – The server acknowledges the request and responds.
- **ACK** – The client acknowledges the server, completing the connection.

This confirms successful TCP communication between both machines.

Screenshot:



8.4 HTTP Analysis

Filter:

http

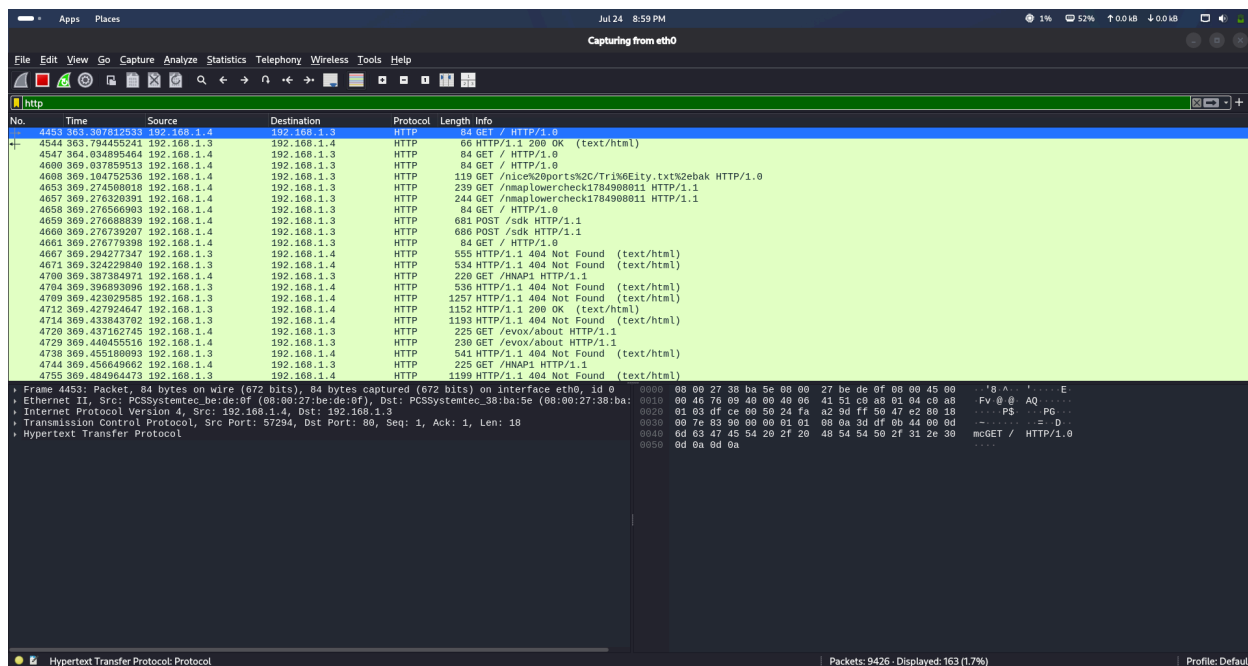
Purpose:

To inspect HTTP communication between the browser and the target web server.

Observation:

HTTP packets were captured when accessing the Metasploitable web application through Firefox. The captured traffic contained HTTP requests and responses exchanged between the client and the Apache web server running on the target machine.

Screenshot:



8.5 FTP Analysis

Filter:

ftp

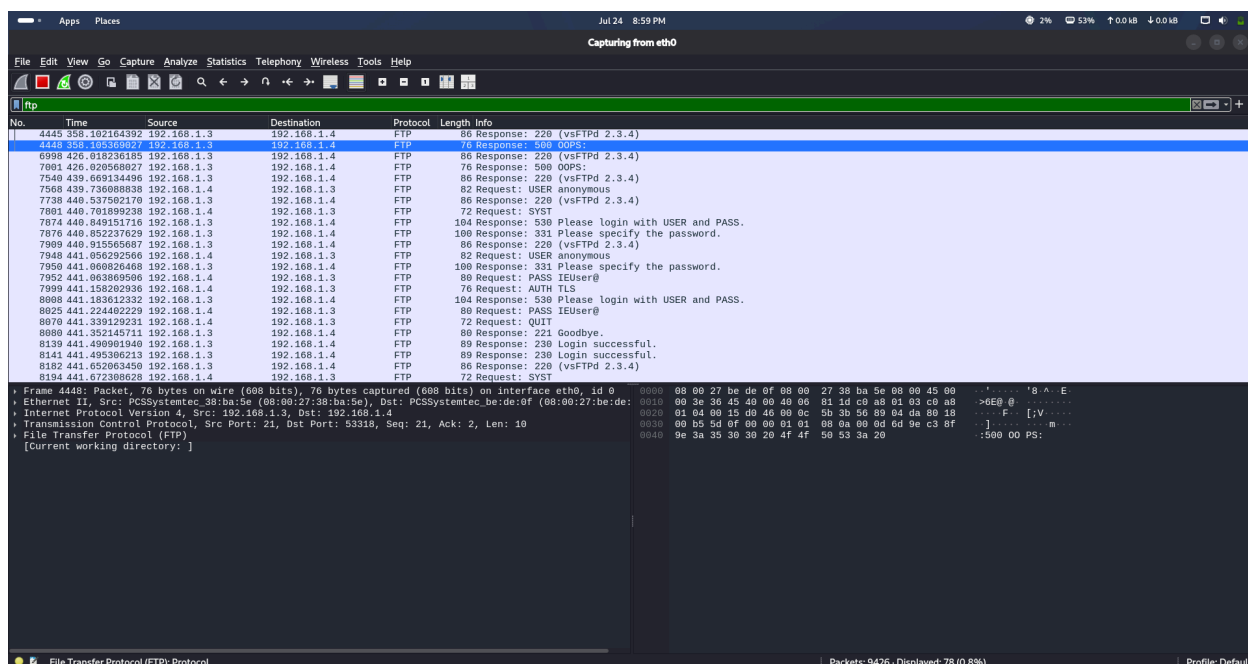
Purpose:

To analyze File Transfer Protocol (FTP) communication between the client and the target server.

Observation:

FTP traffic was captured during the scan. The analysis confirmed that the FTP service was active on the target system. The Nmap scan also indicated that anonymous FTP login was allowed, which may pose a security risk if not properly configured.

Screenshot:



8.6 Telnet Analysis

Filter:

telnet

or

tcp.port == 23

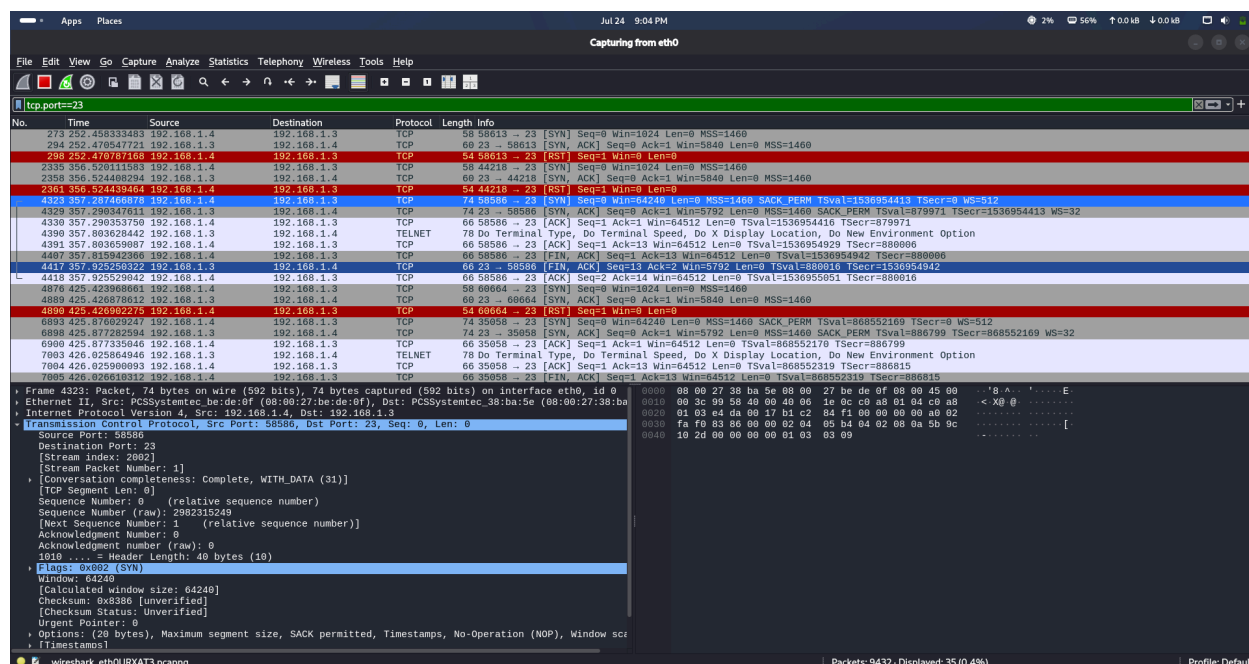
Purpose:

To observe Telnet communication between the client and the target system.

Observation:

Telnet packets were captured, confirming that the Telnet service was running on the target machine. Since Telnet transmits data in plain text without encryption, it is considered an insecure protocol and should be replaced with secure alternatives such as SSH.

Screenshot:



8.7 DNS Analysis

Filter:

dns

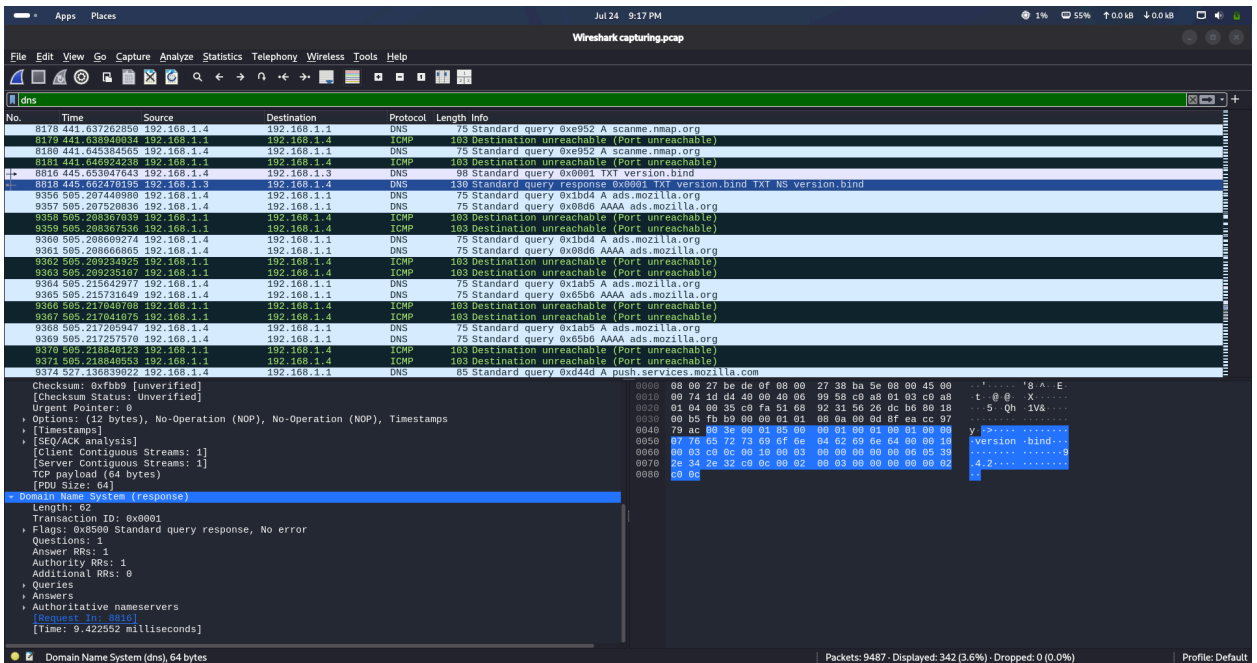
Purpose:

To analyze DNS queries and responses generated during network communication.

Observation:

DNS packets were captured during the assessment. These packets show how domain names are translated into IP addresses before establishing communication with the target or external resources. DNS analysis helps understand the name resolution process within the network.

Screenshot:

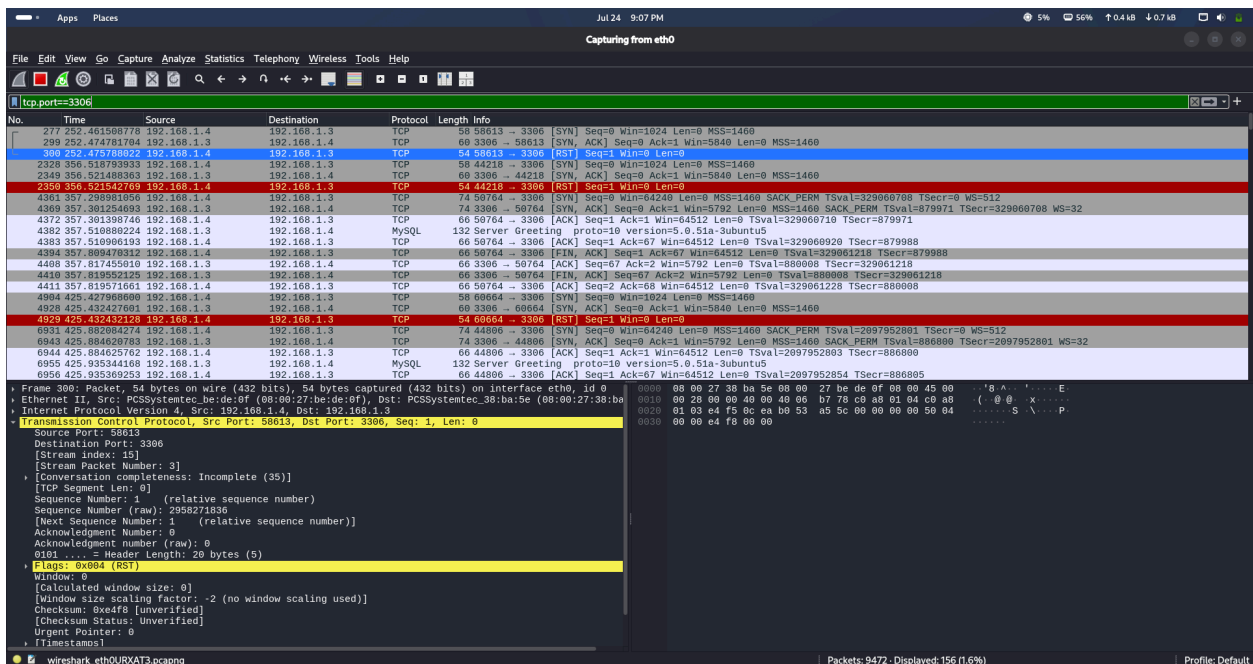


8.8 MYSQL

Filter:

tcp.port==3306

Screenshot:



9. Findings

The following security findings were identified during the penetration testing assessment.

Finding	Severity	Description
Anonymous FTP Login	High	The FTP service allows anonymous login, which may lead to unauthorized access.
Telnet Enabled	High	Telnet service is enabled and transmits data in plain text without encryption.
Outdated Apache Server	Medium	Apache HTTP Server 2.2.8 is outdated and may contain known vulnerabilities.
SMB Signing Disabled	Medium	SMB message signing is disabled, making the service vulnerable to certain network attacks.
Multiple Database Services	Medium	Both MySQL and PostgreSQL services are publicly accessible on the target machine.
Expired SSL Certificates	Low	SSL certificates detected during the scan are expired and should be replaced.
Apache Tomcat Server	Informational	Apache Tomcat 5.5 service was detected on port 8180.

Multiple Open Ports	Medium	Several network services were found running, increasing the target's attack surface.
Outdated SSH Service	Medium	OpenSSH 4.7p1 is an old version that may contain known security issues.

10. Recommendations

Based on the findings, the following security recommendations are suggested:

- Disable anonymous FTP access unless it is specifically required.
- Replace Telnet with SSH for secure remote administration.
- Update Apache, OpenSSH, Samba, MySQL, PostgreSQL, and other outdated services to the latest supported versions.
- Enable SMB message signing to improve communication security.
- Replace expired SSL certificates with valid and trusted certificates.
- Disable unnecessary network services and close unused ports.
- Apply regular security patches and system updates.
- Perform periodic vulnerability assessments and penetration testing to identify newly introduced security risks.

References

1. Nmap Project. *Nmap Reference Guide*. <https://nmap.org/docs.html>
2. Wireshark Foundation. *Wireshark User Guide*. <https://www.wireshark.org/docs/>
3. Kali Linux Documentation. <https://www.kali.org/docs/>
4. Rapid7. *Metasploitable 2 Documentation*. <https://docs.rapid7.com/metasploit/metasploitable-2/>